

PONCA CITY, OK, USA

WMO: 724530

Lat: 36.737N

Lon: 97.102W

Elev: 305

StdP: 97.72

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13969

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.8	-9.0	-18.1	0.8	-9.4	-15.7	1.0	-7.2	11.7	3.6	10.7	3.3	3.7	0	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	38.3	23.5	36.8	23.9	35.1	23.9	26.1	33.5	25.4	32.9	24.9	32.2	4.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.6	30.0	23.2	18.6	29.2	22.7	18.0	28.7	82.6	33.5	79.7	32.9	77.0	32.3	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.6	9.1	8.3	-16.4	40.2	3.8	2.2	-19.1	41.8	-21.3	43.1	-23.4	44.3	-26.1	45.9		
(5)			-16.9	27.5	3.6	1.4	-19.5	28.5	-21.6	29.3	-23.6	30.1	-26.2	31.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	2.5	4.7	10.0	15.3	20.5	25.7	28.3	27.6	23.1	16.4	9.6	3.7		
	DBStd	10.37	5.98	6.48	6.16	5.10	4.47	3.29	3.19	3.34	4.50	5.22	5.46	5.70		
	HDD10.0	786	239	169	76	13	1	0	0	0	0	9	72	206		
	HDD18.3	2115	489	382	264	119	30	1	0	0	11	100	266	454		
	CDD10.0	2858	8	21	77	171	327	472	567	546	393	206	60	11		
	CDD18.3	1146	0	1	6	27	98	223	308	287	154	39	3	0		
	CDH23.3	13291	1	8	63	267	931	2544	3957	3507	1632	353	27	0		
	CDH26.7	6486	0	1	11	65	323	1203	2176	1867	736	101	3	0		
(14) Wind	WSAvg	4.1	4.0	4.3	4.7	4.8	4.3	4.2	3.8	3.6	3.7	3.9	4.1	3.9		
(15) Precipitation	PrecAvg	903	32	32	62	80	120	116	92	90	102	70	60	39		
	PrecMax	1463	123	112	204	169	268	310	283	302	208	329	308	136		
	PrecMin	476	0	0	1	12	18	12	1	4	2	5	0	0		
	PrecStd	231	28	29	49	38	66	62	70	63	57	61	68	34		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.6	24.4	28.0	31.3	34.1	37.9	41.0	40.5	38.8	32.2	26.3	21.2		
		MCWB	12.1	14.2	16.7	19.0	21.8	23.8	23.2	23.4	22.7	20.5	16.7	14.1		
	2%	DB	17.9	21.2	25.0	28.5	32.0	35.8	38.5	38.0	34.9	29.4	23.4	17.9		
		MCWB	10.6	12.4	15.4	17.8	22.1	23.8	23.6	23.5	23.1	19.8	15.4	11.8		
	5%	DB	15.1	18.0	22.6	26.5	30.1	34.0	37.0	36.4	32.9	27.3	21.2	15.2		
		MCWB	9.3	11.4	14.6	17.0	21.3	23.6	24.3	23.7	22.6	18.6	14.6	10.1		
	10%	DB	12.2	15.3	20.1	24.3	28.2	32.5	35.1	34.6	31.0	25.0	18.7	12.8		
		MCWB	7.1	9.5	13.4	16.5	20.7	23.3	24.3	23.8	21.9	17.9	13.0	8.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.5	16.3	19.2	21.3	24.7	26.7	27.2	26.8	25.4	23.0	19.3	16.7		
		MCDB	18.4	20.2	24.8	27.1	30.8	34.1	34.2	33.9	33.4	28.6	23.2	19.7		
	2%	WB	11.9	14.5	17.6	20.1	23.5	25.6	26.2	25.7	24.2	21.8	17.1	13.5		
		MCDB	16.5	19.2	22.2	25.5	29.6	32.8	33.6	33.2	31.7	26.6	21.2	16.5		
	5%	WB	9.4	12.3	16.1	19.0	22.6	24.9	25.5	25.1	23.5	20.6	15.7	10.8		
		MCDB	14.6	17.4	20.9	24.5	28.3	31.6	33.2	32.5	30.5	25.0	19.6	14.4		
	10%	WB	7.3	9.7	14.3	17.7	21.7	24.1	25.0	24.5	22.8	19.1	14.0	8.4		
		MCDB	11.7	15.1	19.6	23.0	27.0	30.4	32.6	31.7	29.2	23.6	18.2	12.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.8	12.4	12.8	12.6	11.5	11.1	11.4	11.6	12.1	12.5	12.4	11.2		
		MCDBR	16.6	16.5	15.9	14.9	12.8	12.4	13.3	13.4	13.1	14.2	15.1	15.0		
	5% WB	MCWBR	10.2	10.0	8.8	7.8	5.6	4.4	3.7	4.0	4.8	6.7	8.7	9.7		
		MCWBR	14.4	14.4	12.9	12.5	11.0	10.9	11.5	11.2	11.2	10.9	12.3	12.9		
(40) Clear-Sky Solar Irradiance	taub		0.292	0.301	0.340	0.381	0.420	0.428	0.435	0.436	0.392	0.345	0.306	0.285		
	taud		2.497	2.485	2.401	2.309	2.272	2.285	2.303	2.245	2.351	2.471	2.512	2.527		
	Ebn at Noon		908	941	926	900	866	856	847	838	860	877	882	894		
	Edh at Noon		80	92	110	128	135	133	130	134	115	93	79	73		
(44) All-Sky Solar Radiation	RadAvg		2.62	3.29	4.29	5.36	5.94	6.78	6.72	6.06	5.06	3.78	2.80	2.22		
	RadStd		0.26	0.49	0.28	0.36	0.57	0.55	0.39	0.36	0.40	0.45	0.33	0.22		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page